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Project Name

Topic area: Emergent Behavior

For my project, I want to build a simulator of atomic interactions representing a simplified nuclear reactor. This will involve modeling the subatomic particles (neutrons/electrons/protons), as well as basic atoms. The interactions will be approximations based on real physics for different types of collision and absorption reactions that can occur. This fits the emergent behavior theme, because while the individual reactions are relatively straightforward and there is only a limited number of possible outcomes when particles interact, the overall behavior of a system with multiple particles can become complex and unexpected.

I will primarily be focusing on the neutron interactions that occur, and not necessarily trying to fully capture all possible reactions and behavior.

Some of the basic rules of behavior will be the following:

- elastic scattering
  - nucleus is struck by a neutron and rebounds, with no absorption occurring
  - transfers kinetic energy to the nucleus, but does not otherwise alter it
- inelastic scattering
  - nucleus absorbs a neutron, emitting a lower enery neutron, then the nucleus de-excites by emitting a photon
  - nucleus is briefly excited and gains a neutron, but emits a neutron and photon to return to its original state and energy level
- radiative capture
  - nucleus absorbs a neutron, then de-excites by emitting a photon
  - this ultimately adds a neutron to the nucleus, creating a heavier isotope
  - occurs when the incoming neutron is lower energy than would be required to cause inelastic scattering - the absorbed energy is not sufficient for the nucleus to emit a neutron
- induced fission
  - a heavy nucleus absorbs a neutron and is excited above the point of stability. It splits into 2 or more smaller nuclei and emits a large burst of neutrons
  - the fission fragments created by this slowly decay to stability and emit more neutrons as they do so

My plan is to use smaller thresholds for the line of stability than exist in real life. For example, maybe making a nucleus become unstable with only 5 or 10 neutrons and a high neutron to proton ratio, instead of only very heavy elements like U-235 fissioning relatively easily. This will be a small scale simulation of atomic and neutron interactions that, depending on the initial particles present will gain energy, absorb neutrons, fission, and potentially create self-sustaining reactions. I'll probably plan to ignore other types of radiation like beta and alpha, and it will just be a small scale system focused on the neutron interactions.

Using simple ascii art will likely be the simplest way to represent the interactions, but if possible I would like to look into the possibilities that exist for graphics libraries in Rust and make a nice visualization if time allows.